AirGuard India: Station-Calibrated Day-Ahead Air Quality Forecasting with Uncertainty-Aware Vulnerability Advisories

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***Abstract*—This development manuscript documents an India-specific station-calibration protocol locked before model fitting; the manuscript itself is not a new preregistration. The intended comparison is raw operational CAMS versus direct learning and residual correction, followed by uncertainty calibration and exposure advisory evaluation. Hourly quality rules, chronological and spatial validation, and a sealed final test constrain the analysis. The genuine historical CAMS issue-time archive has passed operational GRIB validation; development acquisition is incomplete. A separate preliminary measured-data study is reported as infrastructure evidence, without a CAMS improvement claim. Real prospective forecasts are frozen, but matched measured outcomes remain pending. [RESULT PENDING CAMS ARCHIVE EXPERIMENT]**

***Index Terms*—CAMS, air quality, India, calibration, prospective verification**

I. Introduction

A. Study Status

Two streams are kept separate: frozen current forecasts constitute the prospective study; measured historical targets support non-CAMS preliminary research. Neither substitutes for the blocked primary archive comparison.

Station-specific CAMS post-processing is established. Riccio and Chianese study statistical correction with spatial verification [1]. Petetin et al. explicitly distinguish improvements in continuous forecast errors from categorical ozone-warning skill [2]. Operational regional CAMS production also includes MOS [3]. The proposed contribution is an audited Indian evaluation joining issue-time availability, censored-Severe diagnostics and prospective station verification; neither a new MOS invention nor a world's-first application is claimed.

1) Scope of the contribution: For station s and target date d, the predictor $X_{s,d}$ is formed before the declared issue time, while $y_{s,d}$ is constructed later from quality-controlled measurements. The forecast task is next-day prediction, not concurrent regression on target-day pollutant observations. A high-quality interface alone cannot establish correction skill. This paper therefore binds each proposed claim to an immutable result or an explicit pending gate. The released operational product remains a raw CAMS estimate; offline station-correction development is a separate research stream.

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Development manuscript. Primary CAMS results and model selection are pending; the final evaluation interval remains sealed.

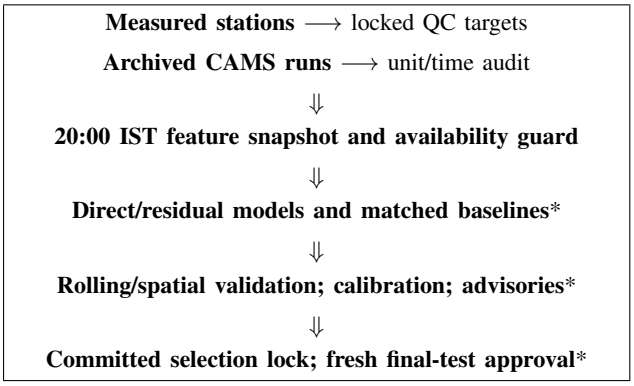


Fig. 1. Study architecture. Asterisks mark unfinished primary experiment stages. Measured outcomes and forecast fields have separate provenance; final evaluation remains sealed.

II. Related Work

A. Problem and Related Work

The research question is whether a correction learned at Indian stations improves next-day CPCB-style AQI while retaining useful uncertainty and warning behaviour. Atmospheric forecasts and measured concentrations have different provenance. Existing Indian forecasting systems establish that this is an evaluation problem, not an unoccupied application domain [4]. Station MOS, CAMS statistical post-processing, conformal intervals and risk calibration are established methods. The targeted review records overlapping European CAMS work and distinguishes Indian reanalysis mapping from genuine issue-time prediction. No world's-first claim is made. The contribution under test is the audited Indian operational procedure and its empirical warning trade-offs. Full-text novelty review remains incomplete.

III. Data and Methods

A. Data, Targets and Quality

OpenAQ preserves upstream CPCB-provider station identity; its measurements are not an official CPCB bulletin or licensed CCR export [5]. The inherited catalogue contains 481 eligible stations across 255 parsed cities; the metadata-only stratified sample selects 181 stations in 100 cities. Current catalogue selection can introduce survivorship bias and is disclosed. The locked v3.1 window is 2025-02-19 through 2026-10-07. The protocol was locked at 2026-10-11T18:32:05.939254+00:00; its prior version and hash remain

in the lock history. Its immutable hash is recorded in the result artifact. This acquisition checkpoint has 179 usable stations, 100 cities, 64799 valid targets and 42481 retained rejected station-days. These are partial-collection counts, not national study acceptance.

The measured semantic dataset and preliminary input share SHA-256 prefix `4761a5da60dde059`; the immutable preliminary run has prefix `287c87db04a6b294`. Full hashes, denominators and artifact locations are recorded in the claim-evidence manifest.

The single CPCB-style engine applies pollutant breakpoints, at least sixteen distinct hours per pollutant, at least three supported pollutants and a required particulate species [6]. Particulates and the designated gases use 24-hour means; CO and ozone use the maximum complete trailing eight-hour mean. Targets are labelled AirGuard CPCB-style AQI. Exact zeros, provider flags, poor coverage and six-hour identical runs are rejected. PM2.5 above PM10 and potential spikes are flagged and retained. Raw caches and row-addressable rejection ledgers preserve exclusions. Unsupported ppb conversion is refused.

Severe AQI is an open lower bound of at least 401. The checkpoint contains 383 censored Severe days. They cannot supply exact residual labels or identified MAE, RMSE, bias or coefficient-of-determination values. Category, exceedance, forecasts below 401 and lower-bound shortfall remain estimable.

1) Concentration-to-index transformation: Within a bounded CPCB interval $[C_L, C_H]$ with index endpoints $[I_L, I_H]$, the pollutant sub-index is

$$I_p(C) = I_L + \frac{I_H - I_L}{C_H - C_L}(C - C_L), \quad I = \max_{p \in S} I_p. \tag{1}$$

The daily support S requires at least three sufficiently covered species, including particulate matter. It is recorded as an outcome annotation, never used to select predictors retrospectively. In the current valid development join, S equals the prerequest frozen PM2.5/PM10/O3 support. Rejected rows retain reasons and provenance rather than disappearing from denominator counts. An open concentration interval does not justify linear extrapolation to a fictitious exact Severe target.

2) Quality-control interpretation: The six-hour stuck-run rule rejects the complete identical sequence, not only its final bin. Exact-zero rejection cannot distinguish a genuinely tiny ozone concentration from a telemetry zero and is therefore a disclosed preregistered trade-off. Conversely, automatic spike deletion could erase real pollution episodes. PM2.5/PM10 inconsistencies and potential spikes remain flagged observations. Sensor rejection, unavailable target and model error are distinct outcomes; their counts are reported separately.

B. Issue-Time Availability

Issue time is 20:00 IST on the preceding day. Observation bins must end by 17:00 IST; daily history is D-2 or earlier. Historical OpenAQ publication time is unknown, so measured-history analyses explicitly assume a healthy feed. Archived weather uses the documented 48-hour Previous Runs offset at https://open-meteo.com/en/docs/previous-runs-api. The Air Quality API documentation is a separate source [7]. FIRMS acquisition time alone does not establish release time. CAMS past fields without their producing run are excluded from the primary experiment. Every admitted snapshot passes its guard.

C. Primary Forecast and Correction

The genuine CAMS global forecast archive [8] must retain initialisation, lead, valid time, units and evidence of pre-issue availability. Reanalysis is prohibited as a replacement. The locked v3.1 protocol specifies target-support-matched CAMS scoring and a six-pollutant secondary comparator. Later target support is an evaluation annotation and cannot become a predictor. The partial join contains 7,560 station-days, 5,506 valid targets and 2,054 retained rejected targets across 42 dates. All 7,560 rows pass the primary availability guard. Every valid target in this subset uses the frozen PM2.5/PM10/O3 support; outcome support remains an evaluation annotation and is excluded from predictors. Full feature integration and the primary runner remain incomplete. The locked comparison is preserved. No primary model has been fitted and no final-test outcomes have been accessed. All methods share observations and evaluation windows. With frozen CAMS index $b_{s,d}$ and exact measured index $y_{s,d}$, residual learning uses

$$r_{s,d} = y_{s,d} - b_{s,d}, \qquad \widehat{y}_{s,d} = b_{s,d} + \widehat{r}_{s,d}. \tag{2}$$

The ladder comprises raw CAMS, direct ML, residual Ridge, residual LightGBM and residual XGBoost, with persistence and prior-only climatology as operational floors. There is no assumption that correction wins. The legacy city model has a different period, target and input representation, and is reported separately.

TABLE I
PRIMARY COMPARISON CONTRACT

Method	Result
Raw operational CAMS	Pending archive
Direct ML	Pending archive
Residual Ridge	Pending archive
Residual LightGBM	Pending archive
Residual XGBoost	Pending archive

1) Verified operational archive checkpoint: The single accepted probe is a genuine forecast (GRIB type fc), initialized 2025-02-18 00:00 UTC with a 24-hour lead and valid at 2025-02-19 00:00 UTC. Its 134-byte PM2.5 field contains nine finite values on a 0.4-degree grid. The development manifest currently verifies 6 of 36 chunks. Requests cover only the frozen station bounding box and development dates. Initialization plus lead reconstructs valid time in every admitted field. Surface PM mass concentrations use 10^9 to convert kg m^{-3} to μg m^{-3}. Gas mixing ratios use the same-run lowest-level temperature T, specific humidity q and hybrid pressure p:

$$\rho = \frac{p}{287.05T(1 + 0.6078q)}, \quad C_{\mu g/m^3} = 10^9 \rho x. \tag{3}$$

CO instead uses 10^6 for mg m^{-3}. Model-level fields are linearly interpolated within the same producing run before density conversion; no extrapolation, reanalysis or post-issue

run is admitted. Seven preceding hourly bins support trailing eight-hour CO/O3 windows. Historical per-run delivery logs are unavailable: pre-issue availability uses the documented nominal dissemination bound, explicitly an assumption. The later ADS retrieval timestamp is provenance, not historical publication.

D. Selection and Ablation

Model settings use training-only inner validation. The locked protocol fixes 90 training days initially, 30 calibration days, 30 test days, a 30-day step and one-day embargo gaps, with expanding training windows. Each fold is reported. The latest 90 dates, 2026-07-10 through 2026-10-07, are sealed. The referenced artifact declares no final-test scoring. The primary selection rule is lowest mean rolling-fold MAE, breaking ties toward Ridge and residual formulations; raw CAMS is retained if no learned method beats it. A committed selection lock, complete isolation audit and fresh user approval are required before final evaluation. Feature ablation separates CAMS, measured history, forecast weather, calendar, spatial context and optional fires. Fire admission requires a date-block interval supporting its benefit. An additional sliding 90-day non-CAMS sensitivity is labelled separately. [RESULT PENDING CAMS ARCHIVE EXPERIMENT]

E. Temporal and Spatial Verification

Primary evaluation will report exact-target MAE, RMSE, signed forecast bias, coefficient of determination, six-category diagnostics, threshold detection, false alarms and censored-Severe shortfall. Poor, Very Poor and Severe events receive explicit diagnostics. Seasonal and regional groups below thirty pairs are listed with counts and withheld metrics. Spatial tests isolate cities, 25-km connected urban clusters, stations and regions. Local-history and monitor-free inference are distinct conditions. Held-out station targets cannot determine training eligibility. The all-station reference is scored on the identical held-out test rows.

Paired comparison uses seven-day moving date blocks with 5,000 resamples and a two-sided Diebold–Mariano daily loss comparison with Newey–West variance and Holm multiplicity adjustment [9]. A confidence interval containing zero does not establish improvement.

F. Uncertainty Calibration

Split conformal intervals at 80% and 90%, quantile calibration and forecast-category Mondrian calibration use only the preceding calibration window. Coverage and widths are reported by category, season and region. For a censored Severe outcome, an upper bound reaching 401 is merely compatible with the known lower bound; it does not identify true-value coverage. Calibration under temporal and spatial shift is evaluated empirically; exchangeability guarantees are not asserted for dependent station-days [10], [11]. Proper uncertainty scoring requires identified outcomes [12].

Marginal coverage conceals important failures: neither interval covered any of the 35 exact Very Poor outcomes in this first fold. Poor-or-worse coverage was also zero on 48 exact pairs. These are preliminary validation failures, not primary CAMS or final-test findings; they preclude claiming reliable high-pollution uncertainty from average coverage alone.

TABLE II
NON-CAMS PRELIMINARY FIRST-FOLD CALIBRATION FAILURES

Method	Exact coverage	Mean width
80% split	0.855	95.5
90% split	0.944	120.6

G. Vulnerability-Aware Advisory Evaluation

Declared exposure strata include general adults, children, older adults, asthma/COPD, heart conditions, pregnancy and outdoor workers. They are not personal medical-risk predictions. P0 uses the point forecast; P1 the calibrated median; P2 the calibrated 0.75 quantile; P3 the smallest calibration-window candidate achieving the corrected empirical under-warning aim $(k+1)/(n+1)$. The aims are 10% for general adults and 5% for the other declared strata. If no candidate qualifies, the policy reports that fact. Under-warning, over-warning, serious under-warning and high-pollution misses are reported alongside date-block confidence intervals. A lower warning rate does not demonstrate a health benefit. Unsupported cohort coefficients remain unadopted. [RESULT PENDING CAMS ARCHIVE EXPERIMENT]

1) Policy losses and calibration boundary: For ordered observed category c and warning category w, the basic losses are $L_- = \mathbf{1}\{w < c\}$ and $L_+ = \mathbf{1}\{w > c\}$. Serious under-warning, missed Poor-or-worse and missed Very-Poor-or-worse events remain separate diagnostics. Lower under-warning can require more false alarms; neither objective is optimized on a test window. The candidate quantiles are fixed before evaluation. P3 chooses on the calibration window only and then reports its frozen test-window behavior, including failure to find any feasible candidate. Its corrected empirical rate $(k+1)/(n+1)$ is a policy selection statistic, not an individual disease probability or a proven dependent-data risk bound.

2) Finite-sample intervals and failure reporting: For exact calibration residuals $e_i = |y_i - \widehat{y}_i|$, split conformal uses the order statistic with rank $\lceil (n+1)(1-\alpha) \rceil$. The interval is $[\max(0, \widehat{y}-q), \widehat{y}+q]$; missing or inadequate calibration support is reported rather than assigned a radius of zero. Mondrian groups use the forecast category, so observed test categories cannot choose an interval. Quantile crossing is repaired using predictions only. Calibration labels exclude censored Severe values from exact residuals; Severe category/threshold behavior is still reported. Under temporal/spatial dependence, coverage is an empirical finding, not automatically a theorem.

3) Exposure guidance and clinical scope: Declared vulnerability strata set environmental-warning tolerances. Shared forecast outcomes do not constitute separate clinical cohort studies. The system does not predict diagnoses, prescribe medication, estimate personal health probabilities or demonstrate avoided illness. Its interface labels measurement versus

TABLE III
DATASET AUDIT. FULL MEASURED COUNTS ARE INHERITED DESCRIPTIVE METADATA; ONLY DEVELOPMENT ROWS ENTER ALIGNMENT.

Item	Count
Measured stations	180
Usable stations	179
Usable cities	100
Retained station-days	107,280
Valid targets	64,799
Rejected targets	42,481
Censored Severe targets	383
Aligned development rows	7,560
Aligned dates	42
Aligned valid targets	5,506
Aligned rejected targets	2,054
Aligned Severe targets	2

TABLE IV
PREREGISTERED SEARCH SPACE, NOT SELECTED HYPERPARAMETERS. ALL MODEL SELECTION REMAINS PENDING.

Candidate	Training-only choices
Ridge	$\alpha \in \{0.1, 1, 10, 100\}$
LightGBM	Leaves 15/31; minimum child samples 20/60
XGBoost	Depth 4/6; minimum child weight 5/20
Both boosters	Learning rate 0.05; maximum 600 rounds; early stopping 40 rounds; subsample and column fraction 0.8
All learners	Seed 42; direct and residual formulations
Selection	Mean rolling-fold MAE; simpler learner and residual formulation break ties

modelling, observation age, unsupported intervals and raw-model uncertainty. Clinical coefficients remain unvalidated.

4) Baseline estimation and operational missingness: Climatology is estimated from exact training targets in a station's calendar month, with station and global means as explicit fallbacks. These statistics are re-estimated inside each training fold. Persistence uses the measured rolling 24-hour index ending at the cutoff, rather than the not-yet-complete day immediately before the target. Consequently a persistence forecast can be missing even when the eventual target is valid. The primary comparison requires exact targets, the CAMS baseline and persistence jointly; the secondary set relaxes only the persistence requirement. Reporting these denominators prevents a method gaining an advantage by being evaluated on easier, more complete days.

The implementation joins on station identifier and calendar date and rejects duplicate keys or mismatched station coordinates. Calendar lags are looked up by date, rather than shifted across whichever observations happen to be present. An absent day is therefore missing, not replaced by an older reading with the wrong lead. Censored historical values retain a lower-bound indicator; they do not supply fabricated exact residual-bias labels. Numerical missingness is handled by training-only imputation with indicators. This choice describes the current implementation and remains subject to the development audit; no imputer is fitted to a validation or final window.

Nearby-station summaries use only donor rolling measurements whose bins ended by the cutoff. Self-donations are excluded. Spatial inference without a local monitor must mask unavailable donor stations as well as the held-out station's own history. A held-out target is never a contemporaneous neighbour feature. Elevation is static cached terrain metadata, not an outcome-derived station-selection criterion. The acquired surface cube also provides maxima, ranges, standard deviations and daytime-minus-nighttime concentrations; all come from the same producing run as the daily CAMS baseline.

5) Scope of the operational reconstruction: The locked YAML preserves the earlier proxy-source disclosure and its hash; it has not been silently rewritten to claim historical operational access. The separately frozen acquisition manifest identifies the genuine ADS forecast stream now being integrated. That provenance improvement does not relax the original QC, date splits, censoring or selection criteria. It also does not prove each forecast's historical delivery time. Nominal publication and healthy-feed assumptions remain explicit limitations of retrospective reconstruction, to be checked prospectively.

IV. DEVELOPMENT RESULTS AND PENDING EXPERIMENTS

A. Preliminary Results Contract

The separate non-CAMS run completed 24 temporal evaluations, 36 spatial holdouts and 12 calibration-window evaluations; 0 temporal and 0 spatial entries lacked sufficient data. Its predictions and per-fold metrics come from the exact immutable artifact recorded in the claim-evidence package. Strict Ridge had lower paired MAE than climatology (signed difference -8.96 AQI; 95% date-block interval [-12.49, -5.61]; n=38382 exact pairs); lower MAE supported by this interval. History Ridge had higher paired MAE than persistence (signed difference +0.61 AQI; 95% date-block interval [-0.41, +1.70]; n=31622 exact pairs); a nonzero MAE difference is not established by this interval. These preliminary comparisons do not establish production approval. They do not answer whether CAMS station correction improves a genuine operational baseline. The definitive results table and skill figure remain: [RESULT PENDING CAMS ARCHIVE EXPERIMENT]

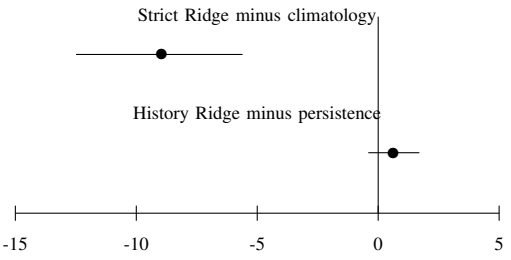


Fig. 2. NON-CAMS preliminary paired MAE differences and 95% seven-day date-block intervals. Zero means equal MAE; lower is better. Strict $n = 38,382$; history $n = 31,622$.

1) Limits of the preliminary experiment: The non-CAMS plan uses 30 valid training days per station and a fixed Ridge penalty of 10; these choices must not be mistaken for primary CAMS tuning. The locked primary station rule remains 120 valid training days. Thus the first 90-day primary

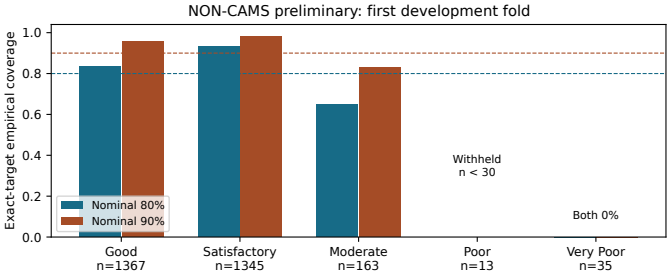


Fig. 3. First-fold non-CAMS split-conformal coverage. Poor has only 13 exact pairs and its subgroup metric is withheld; Very Poor has 35. Neither constitutes primary CAMS evidence.

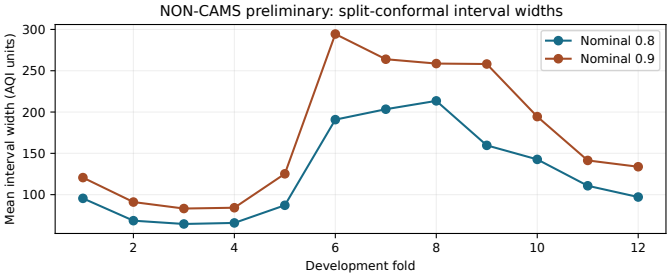


Fig. 4. Mean interval widths across non-CAMS development folds, directly from the immutable result. These are fold summaries, not a manufactured distribution of individual interval widths.

training window cannot meet station eligibility; it must be reported as insufficient, not rescued by future coverage or a silent threshold reduction. Later windows are evaluated when sufficient. Primary model selection has not begun. Absence of a primary score is not evidence of equivalence, and preliminary skill against climatology cannot establish benefit over archived CAMS or persistence.

B. Prospective Verification

The current audited ledger contains 24 real forecasts across 2 target days, and 0 verified days. 0 forecasts have a bound station. 12 historical forecasts lack an issuance Git SHA; missing historical identities will not be invented. New issuance now records its deployment identity. Unbound records cannot be made station forecasts by retrospectively selecting an outcome monitor. No experimental correction is deployed. Issued forecasts remain immutable; a thirty-day track record requires real elapsed days and matched measured outcomes. Station-level prospective evaluation remains blocked until stations are bound before issuance.

1) Interpretation of incomplete primary evidence: Partial forecast extraction is an integrity milestone, not a forecast-skill result. The current matched period cannot satisfy the 120-valid-day station rule or the prescribed calibration and evaluation windows. The pipeline therefore emits explicit insufficient-data entries rather than training on a shorter interval. This affects all primary candidates, including raw CAMS, climatology and persistence, so their intended paired comparison is not quietly replaced by a convenient short-period leaderboard.

The non-CAMS result establishes that the experimental machinery can recover differences against climatology, but the persistence comparison does not establish an advantage for history Ridge. Likewise, near-nominal marginal coverage can coexist with complete failure on a sufficiently supported high-pollution subgroup. Those observations motivate the pre-registered severity and advisory analysis; they do not select a primary model. No primary error scatter plot, error histogram or health-effect plot is supplied because the required primary prediction artifacts do not yet exist.

V. Discussion and Reproducibility

A. Reproducibility

Files and decompressed cache content have SHA-256 manifests. Dataset identity, feature version, protocol hash, source hash and run configuration accompany each result. Local station reconstruction is audited independently of derived caches. A clean-clone measured-dataset reconstruction passed its semantic hash audit under the recorded runtime. A complete primary reproduction remains pending. Numerical values in this skeleton are generated from evidence artifacts, never filled manually.

B. Limitations and Planned Completion

ADS setup is complete. Incomplete archive coverage and primary experiments block the genuine CAMS comparison. Partial station acquisition, current-catalogue survivorship, unknown historical observation latency and limited extreme-event support bound the preliminary analysis. Optional fire release time remains unproven. The initial station-free prospective forecasts do not establish station forecast skill. Final-test isolation, transparent artifacts and explicit missing results are necessary conditions for completion, not evidence of improved accuracy. The intended completed narrative is CAMS baseline, India-specific correction, rolling/spatial testing, calibrated uncertainty, cohort advisory trade-offs and prospective station verification.

1) Artifact lineage and release boundary: The reproducibility package binds the protocol hash, measured semantic hash, operational request manifest, GRIB checksums, derived table hash and feature version. Raw files are stored outside Git, while manifests and implementation are versioned. Resumption polls existing request identifiers before submitting new work. Hash failures stop processing instead of replacing evidence in place. The development reader applies its date exclusion before materialising outcomes. The readiness report itself reads artifact metadata and hashes file bytes; it does not run an estimator or open final labels.

Production validation and scientific validation answer different questions. The released interface has route, API and browser checks, while the proposed station correction still requires completed development experiments. A successful web deployment is not a substitute for predictive validation. Similarly, a compiled manuscript is a presentation artifact and cannot release a research gate. A selected model, if justified later, will need a committed feature and parameter lock, uncertainty and advisory configurations, and a complete evidence audit before final-test authorization is requested.

VI. Conclusion

A. Conclusion

The study preserves its locked protocol, measured targets and immutable preliminary experiment; genuine CAMS alignment is now audited on a partial development subset. Correction performance, operational intervals, advisory risk attainment and thirty-day verification remain hypotheses. [RESULT PENDING CAMS ARCHIVE EXPERIMENT]

Ethics and AI Disclosure

No human-subject data was collected. Cohort guidance is environmental exposure advice, not validated medical advice. Any participant study requires prior ethics approval and consent. Claude Code and OpenAI Codex assisted software, tests and drafting. Authors must review the methods, bibliography, disclosures and generated claims before submission. The named author must approve the final manuscript before submission.

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